



Course Introduction

Prof. Antonino Nocera



Agenda

- Introduction
- Course Plan
- Assessment Description
- General Information



Introduction



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Research Topics:

- Artificial Intelligence
- Security of Artificial Intelligence
- Data Science & Big Data Analytics
- Social Network Analysis
- Data Mining, Text Mining, NLP
- Data Privacy Models
- Cyber Security (Mobile, IoT, Blockchain, Trust and Reputation)

dcalab

Adversarial Attacks

- FGSM
 - Untargeted
 - Targeted



x

“panda”

57.7% confidence

+ .007 ×



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

=



$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

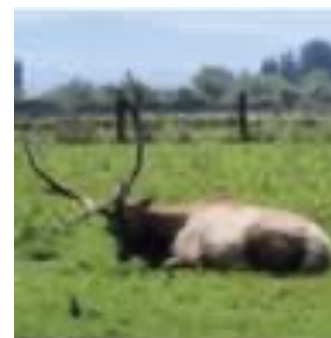
99.3 % confidence





Backdoor Attacks

- BadNets

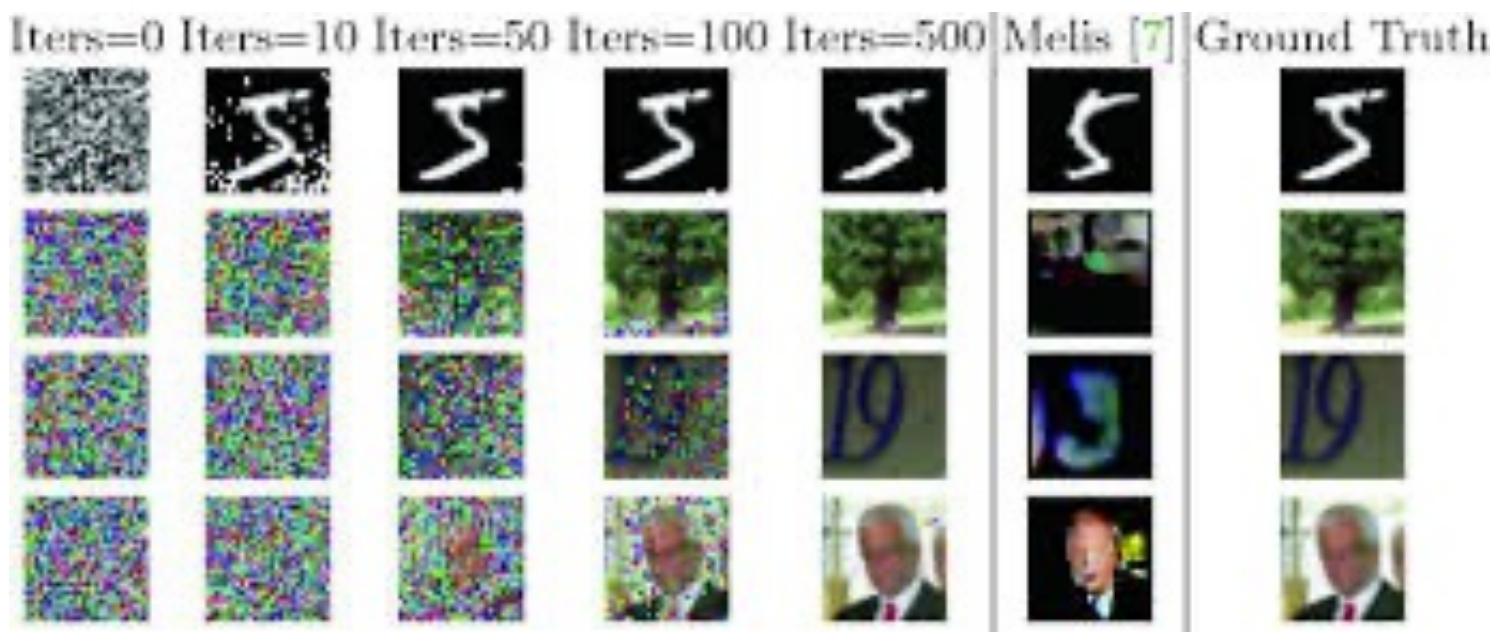


a) clean



b) Badnets

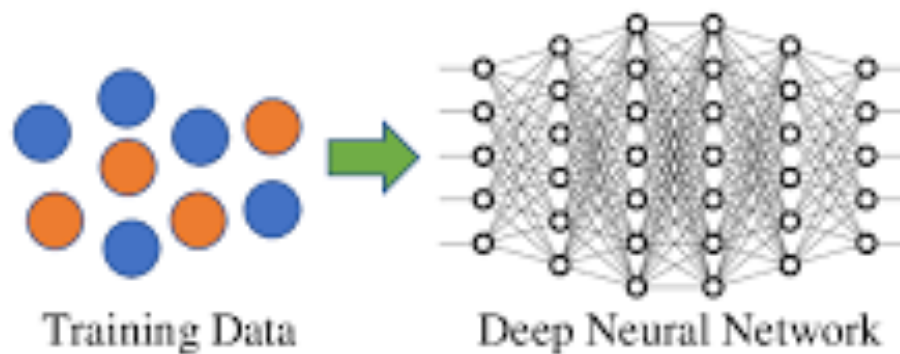
Dataset Leakage From Gradients



Membership Inference Attack



Training of Target Model



Membership Inference Attack on Target Model



Backdoor Defense Example

